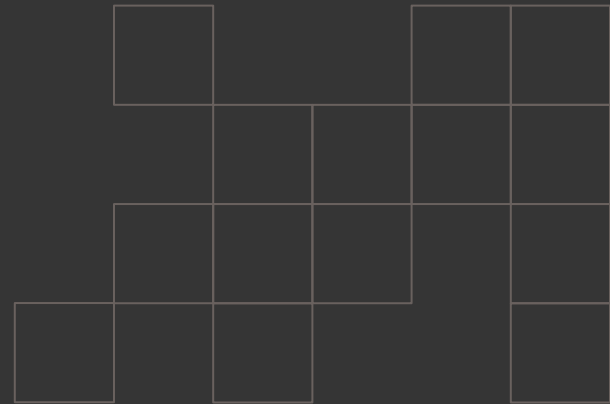


STAT 230
4/30/2025

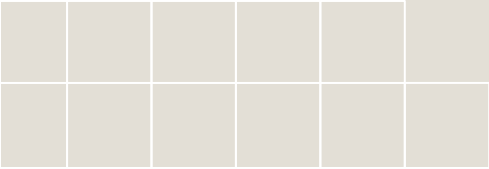
Final Project presentation

By Muhammad Baig



Content

1.
Dataset Overview
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Histogram- Model year
4.
Boxplot- King vs Pierce
5.
Test Results
6.
Conclusion



This Project is about EV population data analysis

Data acquired from data.gov

Dataset: Washington State Electric Vehicle
Registrations

Key variables:

Model Year

Make & Model

County

Electric Range

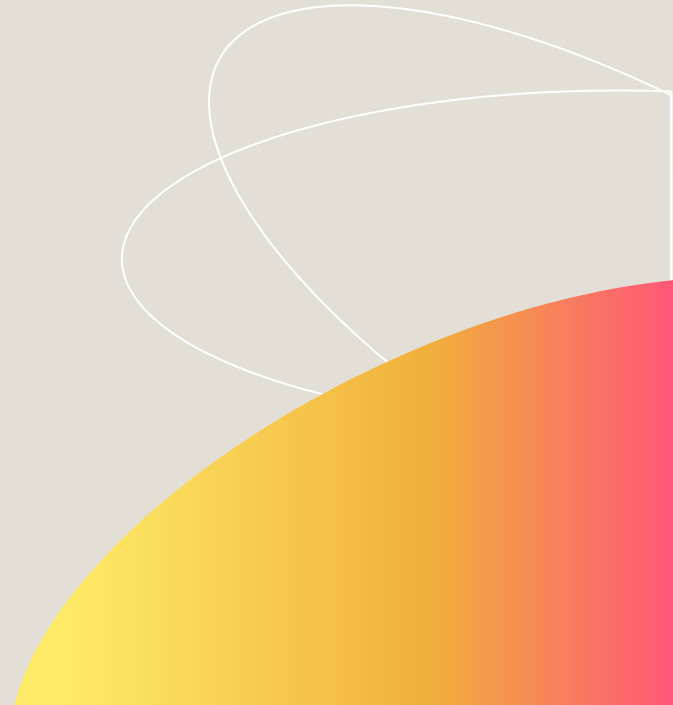
Vehicle Type (BEV, PHEV)



Research Questions:

How has the number of electric vehicles changed across model years?

Is there a significant difference in model years between King County and Pierce County?



Growth of Electric Vehicles by Model Year

1

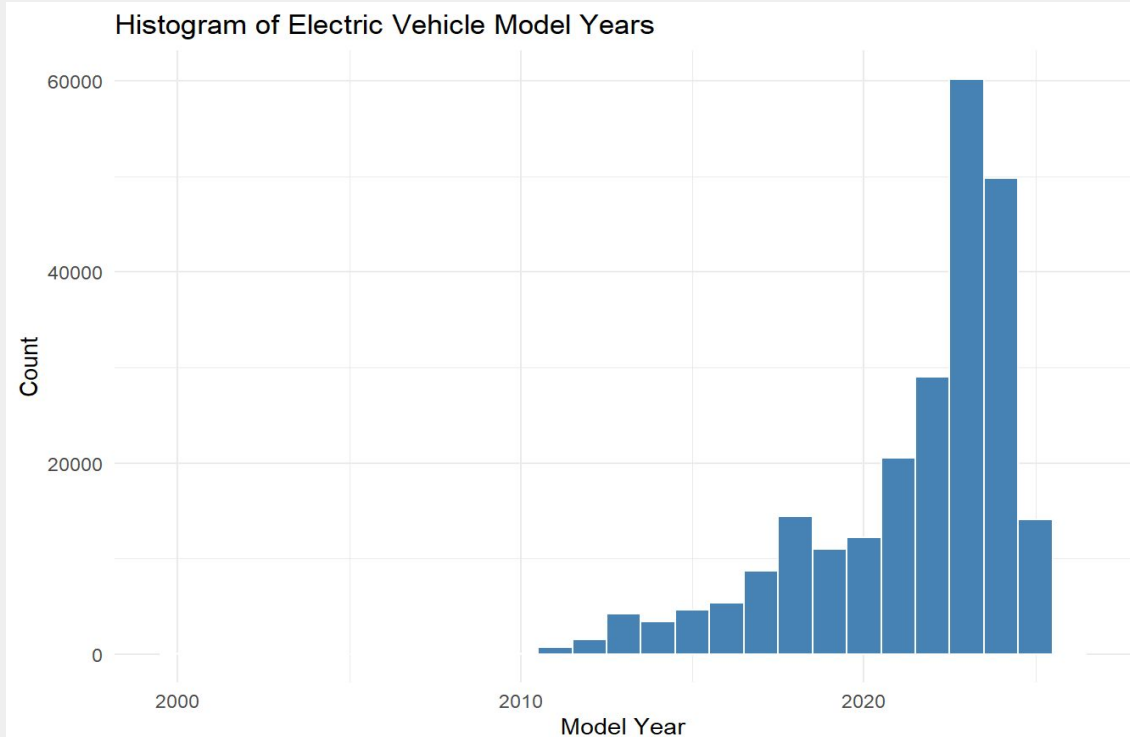
Most vehicles are from 2017-2023

2

Rapid growth in recent years

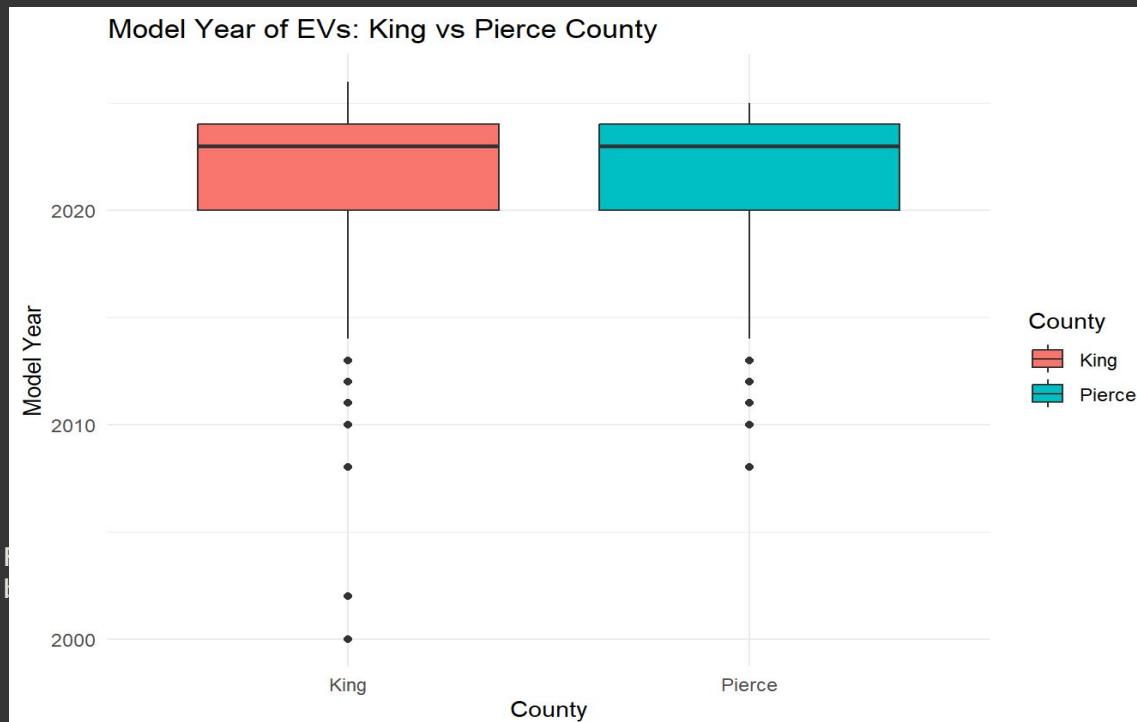
3

EV adoption trend is accelerating



Comparing Model Years by County

1. King County has slightly newer EVs
2. Distribution is similar but shifted slightly higher
3. Useful for visualizing regional trends



Two-Sample T-Test

Null Hypothesis
(H_0): No
difference in
mean model years

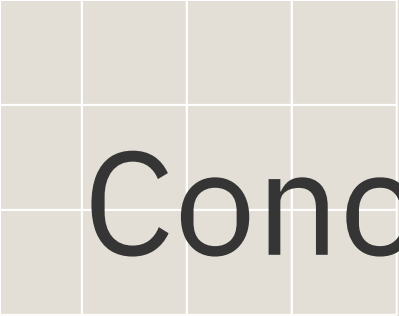
$t = 5.32$, $p\text{-value} \approx$
 $1.03e-07$

Conclusion:
Statistically
significant difference

Notes:

Even a small
difference can be
statistically
meaningful

Supports findings
from earlier
descriptive stats

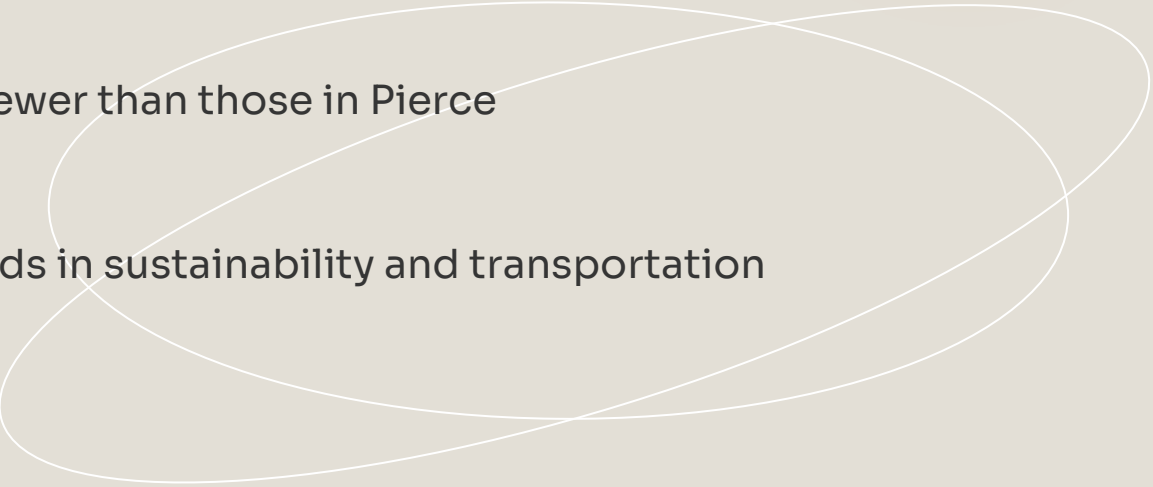


Conclusion

EV adoption is increasing, especially in recent model years

King County EVs are statistically newer than those in Pierce

Public data reveals real-world trends in sustainability and transportation





Thank you